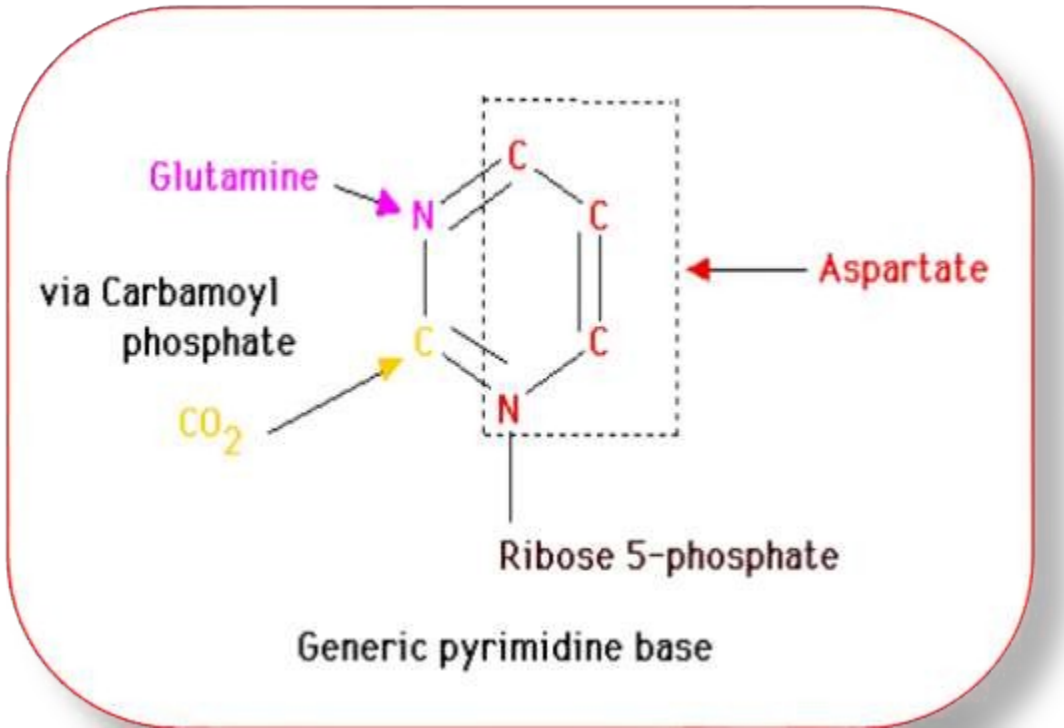
Pyrimidine Synthesis

Pyrimidine

- Cytosine
- Thymine
- Uracil



- **The pyrimidine ring is synthesized as free pyrimidine & it is incorporated into the nucleotide.**
- **Aspartate, glutamine (amide group) & CO_2 contribute to atoms in the formation of pyrimidine ring.**

Step 1: Carbamoyl Phosphate Synthesis

- The reaction occurs in cytoplasm.
- Glutamine transfers its amido nitrogen to CO_2 to produce carbamoyl phosphate.
- This reaction is catalyzed by ATP dependent enzyme carbamoyl phosphate synthetase II (CPS II).
- CPS-I is involved in urea cycle.

Step 2: Rate Limiting Step

- Carbamoyl phosphate condenses with aspartate to form carbamoyl aspartate.
- This reaction is catalysed by aspartate transcarbamoylase.
- The atoms C2 & N3 are derived from carbamoyl phosphate & the rest are from aspartate.

Step 3: Formation of Pyrimidine Ring

- Dihydroorotase catalyses the pyrimidine ring closure with a loss of H_2O.
- Step 4: Oxidation
- NAD^+ dependent dehydrogenation, leading to the formation of orotate.
- The enzyme is dihydro orotate dehydrogenase (DHODH).

Step 5: Formation of OMP

- Ribose 5-phosphate is now added to orotate to produce orotidine monophosphate (OMP).
- This reaction is catalysed by orotate phosphoribosyltransferase.
- PRPP is the donor of ribose 5-phosphate.

Step 6: Decarboxylation

- OMP undergoes decarboxylation to uridine mono-phosphate (UMP).
- The enzyme is OMP decarboxylase (OMPDC).
- This is the first pyrimidine that is synthesized.
- 6-aza-uridine inhibits this step & used as an anticancer drug.

- **Orotate phosphoribosyl transferase & OMP decarboxylase are domains of a single protein.**
- **A defect in this bifunctional enzyme causes orotic aciduria.**

Step 7: Synthesis of Triphosphates

- By an ATP-dependent UMP kinase reaction, UMP is converted to UDP which serves as a precursor for the synthesis of dUMP, dTMP, UTP & CTP.
- UDP is phosphorylated to UTP (uridine triphosphate) with the help of ATP.
- The enzyme is nucleoside diphosphate kinase.

- **Ribonucleotide reductase converts UDP to dUDP by a thioredoxin-dependent reaction.**
- **Thymidylate synthetase catalyses the transfer of a methyl group from N^5, N^{10} - methylene tetrahydrofolate to produce deoxythymidine monophosphate (dTMP).**

Step 8: Formation of CTP

- Cytidine triphosphate (CTP) is synthesized from UTP by amination.
- CTP synthetase is the enzyme & glutamine provides the nitrogen.
- CTP synthetase needs ATP.

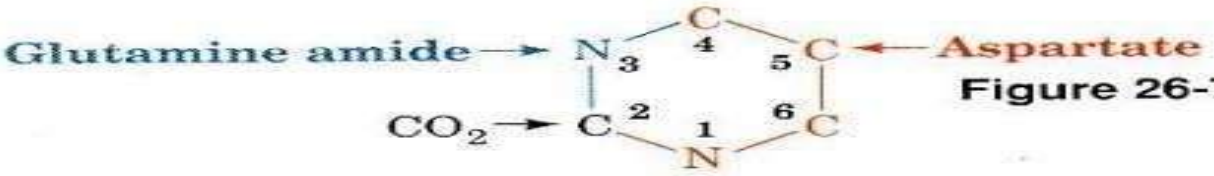


Figure 26-7. The biosynthetic origins of the pyrimidine ring atoms.

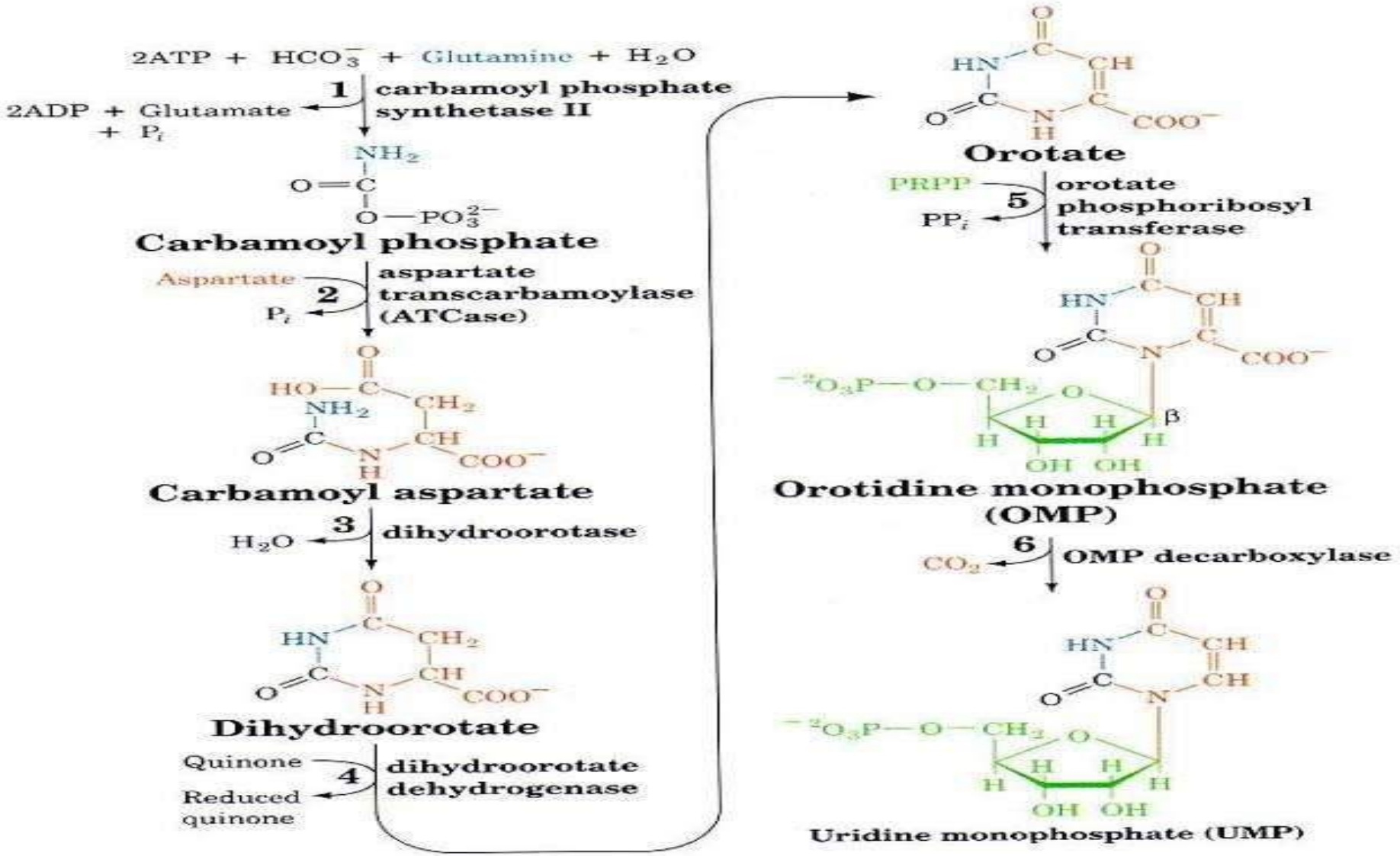
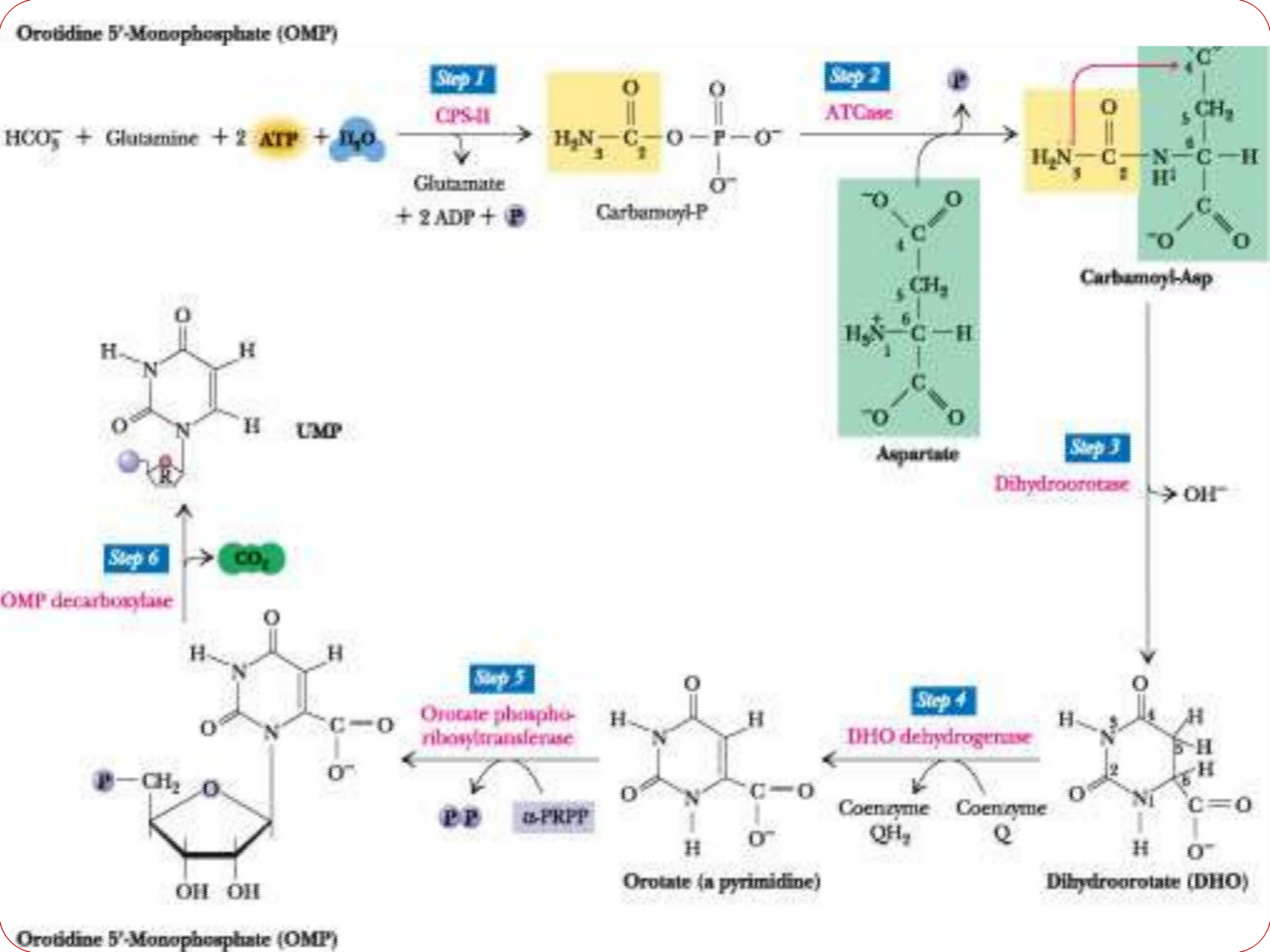
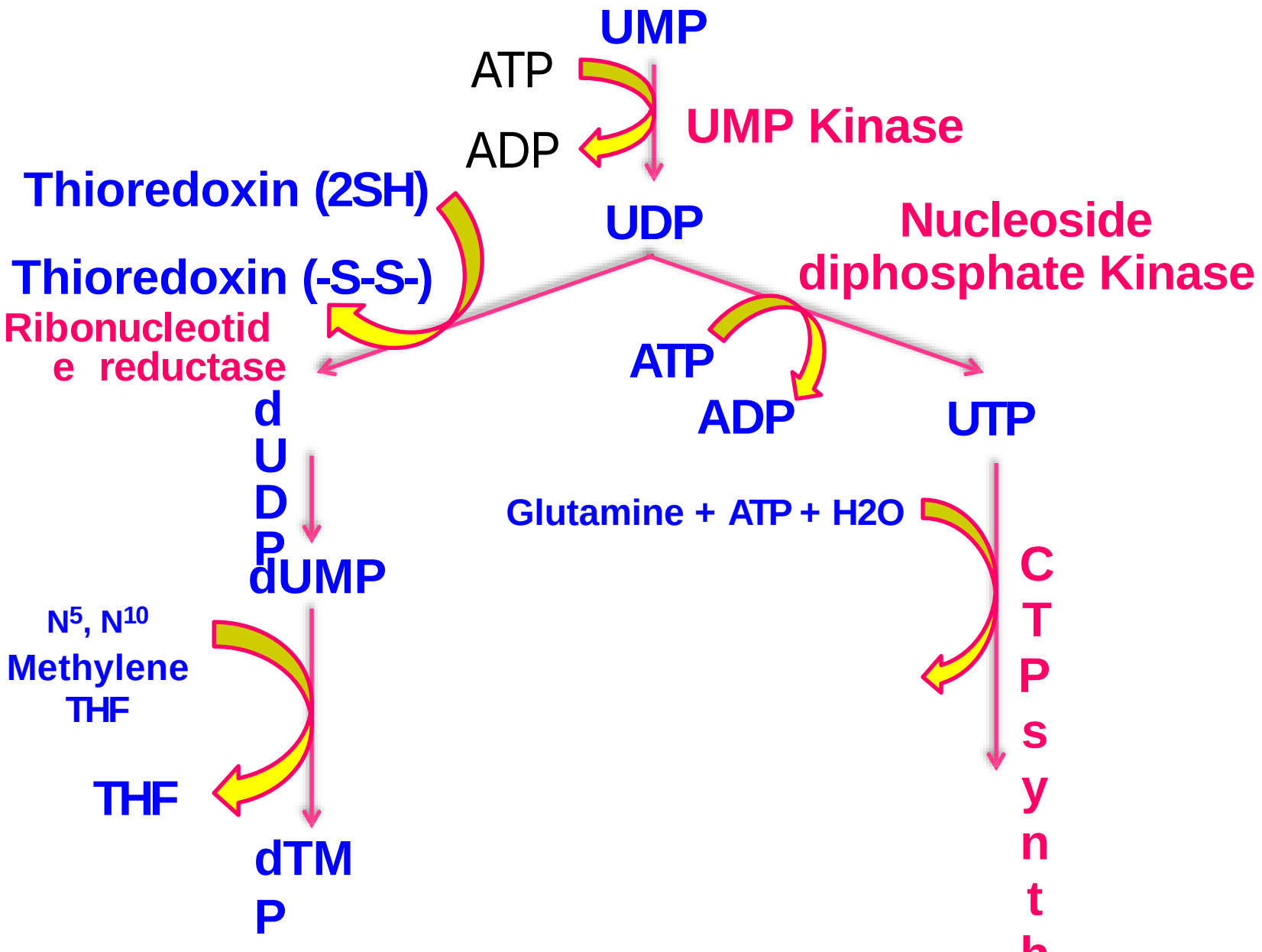


Figure 26-8. The metabolic pathway for the *de novo* synthesis of UMP.


Orotidine 5'-Monophosphate (OMP)
HCO3− + Glutamine + 2 ATP + H2O
Step 1
CPS-II
Glutamate + 2 ADP + P
Carbamoyl-P
Step 2
ATCase
P
Aspartate
Carbamoyl-Asp
Step 3
Dihydroorotase
OH−
Dihydroorotate (DHO)
Step 4
DHO dehydrogenase
Coenzyme QH2
Coenzyme Q
Orotate (a pyrimidine)
Step 5
Orotate phospho-ribosyltransferase
PP
α-PRPP
Step 6
OMP decarboxylase
CO2
UMP
Orotidine 5'-Monophosphate (OMP)


UMP
ATP
ADP
UMP Kinase
Thioredoxin (2SH)
Thioredoxin (-S-S-)
Ribonucleotide reductase
UDP
Nucleoside diphosphate Kinase
ATP
ADP
UTP
dUDP
dUMP
Glutamine + ATP + H2O
CTP synth
N5, N10 Methylene THF
THF
dTMP

Regulation of Pyrimidine synthesis

- In bacteria, aspartate transcarbamoylase (ATCase) catalyses a committed step in pyrimidine biosynthesis.
- ATCase is a good example of an enzyme controlled by feedback mechanism by the end product CTP.

- **Carbamoyl phosphate synthetase II (CPS II) is the regulatory enzyme of pyrimidine synthesis in animals.**
- **It is activated by PRPP and ATP & inhibited by UDP & UTP.**
- **OMP decarboxylase, inhibited by UMP & CMP, also controls pyrimidine formation.**

Pyrimidine Degradation

Pyrimidine Degradation

- The pyrimidine nucleotides undergo similar reactions (dephosphorylation, deamination & cleavage of glycosidic bond) like that of purine nucleotides to liberate the nitrogenous bases cytosine, uracil & thymine.

- The bases are then degraded to highly soluble products β-alanine & β-aminoisobutyrate.
- These are the amino acids which undergo transamination & other reactions to finally produce acetyl CoA & succinyl CoA.

Salvage pathway

- The pyrimidines (like purines) can also serve as precursors in the salvage pathway to be converted to the respective nucleotides.
- This reaction is catalysed by pyrimidine phosphoribosyltransferase which utilizes PRPP as the source of ribose 5-phosphate.

Disorders of pyrimidine metabolism

- **Orotic aciduria:**
- **This is a rare metabolic disorder.**
- **Characterized by the excretion of orotic acid in urine, severe anemia & retarded growth.**
- **Enzyme deficiency: Orotate phosphoribosyl transferase & OMP decarhoxylase of pyrimidine synthesis**

- The condition can be treated by feeding cytidine or uridine.
- They may be converted to UTP which can act as feedback inhibitor.
- Orotic aciduria may also occur in ornithine transcarbamoylase deficiency (urea cycle enzyme) as carbamoyl phosphate accumulates due to defective conversion to citrulline.

Reye's syndrome

- This is considered as a secondary orotic aciduria.
- Due to defect in ornithine transcarbamoylase (of urea cycle) causes the accumulation of carbamoyl phosphate.
- This is then diverted for the increased synthesis & excretion of orotic acid.

References

- Lippincott
- **Chatterjea**